NANDITA KUMARI
317 LINCOLN FIELD BUILDING,
BROWN UNIVERSITY, PROVIDENCE, RI, 02912
WEBSITE

WORK EXPERIENCE

Postdoctoral Researcher, Brown University — SUMMER'24-Present

Instructor, *Summer@Brown, Brown University* — SUMMER'25
Course Name: Travelling through time using the Moon

Mentor, *LUNASCOPE, Brown University* — SUMMER'25

Mentor, *GWISE, Stony Brook University* — FALL'22-SPRING'23

EDUCATION

Stony Brook University — GPA 3.97/4.0 — 2024
PhD in Geosciences and Advanced Certification in Data and Computational Sciences

College of Engineering Guindy, Anna University — GPA 8.93/10 — 2018
Bachelor of Engineering in Geoinformatics

AWARDS

- **LEAG Travel Award**, 2022,2023, 2024
- **Goldschmidt Conference Workshop on Exoplanets** Travel Award, 2024
- **McDonnel Centre for Space Sciences** Travel Award, 2024
- **NASA Exploration Science Forum** Travel Award, 2023
- **Zonta International Amelia Earhart Fellowship,** 2023
- **San Francisco Volcanic Field,** Fieldwork Travel Award, 2022
- **Featured Early Career Presentation**, Lunar Exploration Analysis Group (LEAG) Meeting 2022
- **Best Poster Award** in National Space Science Symposium 2019
- **Mike Barnsley Memorial Award**[1] by Alumni Association of CEG
- **Alumni Association Endowment Award 2007-2**[2] in Geoinformatics Engineering batch 2018
- **Merit Scholarship** from Army Welfare Association

MISSION INVOLVEMENT

- **Science Team Member,** Diviner Lunar Radiometer Experiment

INVITED LECTURES, TALKS AND MEDIA INTERVIEWS

THERMAL INFRARED SPECTROSCOPY, REMOTE SENSING, BROWN UNIVERSITY, 2024

INTREPID AIR AND SPACE MUSEUM, NEW YORK, 2025

DIVINER MEETING, OXFORD UNIVERSITY, 2022

VELAY MEETING, STONY BROOK UNIVERSITY, 2023

CONTRIBUTIONS TO ARTEMIS III, SBU COVERAGE , LONG ISLAND MEDIA

AMELIA EARHART FELLOWSHIP, SBU COVERAGE

PEER REVIEWED PUBLICATIONS

[13] Rogers, A. D., Achilles, C. N., Baldridge, A. M., Honniball, C. I., Knudson, C. A., McAdam, A. C., Henderson, M. J., Hurtado, J., Morse, Z. R., Scheidt, S.P., Whelley, P. L., Young, K. E., Glotch, T. D., Hopkins, R. J., **Kumari, N.,** Raletta, J., Zimmerman, N. A., (2025) Mineralogy and Geochemistry of Xenoliths and Phreatomagmatic Deposits in Portillo Volcanic Field, New Mexico Determined from Portable Instruments: Applications to Surface Exploration of the Moon and Mars. *The Planetary Science Journal, 6(10), p. 235*

[1] Annual award for highest marks in image processing in theory and lab course
[2] Department rank 1 for the academic year 2016-2017
*Equal contribution

[12] **Kumari, N.,** Mustard, J. F., Glotch, T. D. (2025). Particle Size and Albedo Effects on Emissivity Spectra of Lunar Analog Minerals and Rocks in the Intermediate Infrared Region, *Icarus, 441, 116721*

[11] **Kumari, N.,** Breitenfeld, L.B., Shirley, K.A., Glotch, T. D. (2025). Characterizing Extreme Compositions on the Moon Using Thermal Infrared Spectroscopy, *JGR-Planets, 10.1029/2024JE008814*

[10] Noe Dobrea, E.Z., Banks, M.E., Clark, R.N., Wettergreen, D., Hendrix, A., Aherns, C., Bell, E., Breitfeld, A., Bristow, T.F., Buxner, S., Candela, A., Hansen, M., Holsclaw, G., Knightly, P., Kramer, G., **Kumari, N**., Lane, M.D., Martin, A., Meier, M., Patterson, R., Pearson, N., Prettyman, T., Swayze, G.A., Vaniman, D., Vijayarangan, S., Vilas, F., Wright, S.P. (2024). Rover Science Autonomy in Planetary Exploration: Field Analog Tests. *The Planetary Science Journal, 6(2), p. 51.*

[9] **Kumari, N.,** Glotch, T.D., Shirley, K.A., Greenhagen, B.T. and Byron, B.D., (2024). Effects of Space Weathering on the Christiansen Feature Position of Lunar Surface Materials. *Icarus*, *412*, p.115976.

[8] **Kumari, N.,** Glotch, T.D., Williams, J.P., Sullivan, M.T., Li, S., Greenhagen, B.T., Waller, D., Powell, T., Elder, C.M., Byron, B.D. and Shirley, K.A., (2024). Extended Silicic Volcanism in the Gruithuisen Region—Revisiting the Composition and Thermophysical Properties of Gruithuisen Domes on the Moon. *The Planetary Science Journal*, *5*(6), p.132.

[7] Mahanti, P**.,** Williams, J.P., Robinson, M.S., Wagner, R., Mazarico, E., Henriksen, M., Brown, H. and **Kumari, N.**, (2024). Dynamic Secondary Illumination in Permanent Shadows within Artemis III Candidate Landing Regions. *The Planetary Science Journal*, *5*(3), p.62.

[6] **Kumari, N.,** Bretzfelder, J.M., Ganesh, I., Lang, A. and Kring, D.A., (2022). Surface conditions and resource accessibility at potential Artemis landing sites 007 and 011. *The Planetary Science Journal*, *3*(9), p.224.

[5] Williams, J.P., Greenhagen, B.T., Bennett, K.A., Paige, D.A., **Kumari, N.,** Ahrens, C.J., Rubanenko, L., Powell, T.M., Prem, P., Blewett, D.T. and Russell, P.S., 2022. Temperatures of the Lacus Mortis region of the Moon. *Earth and Space Science*, *9*(2), p.e2021EA001966.

[4] Glotch, T.D., Jawin, E.R., Greenhagen, B.T., Cahill, J.T., Lawrence, D.J., Watkins, R.N., Moriarty, D.P., **Kumari, N.,** Li, S., Lucey, P.G. and Siegler, M.A., (2021). The Scientific Value of a Sustained Exploration Program at the Aristarchus Plateau. *The Planetary Science Journal*, *2*(4), p.136.

[3] Thompson, M., Barnes, J., Blewett, D., Cahill, J., Denevi, B., Donaldson Hanna, K., Gillis-Davis, J., Glotch, T., Jha, D., Kramer, G. and **Kumari, N.,** (2021). Space weathering Across the Solar System: Lessons from the Moon and Outstanding Questions. *Bulletin of the American Astronomical Society*, *53*(4), p.172.

[2] Hurley, D., Blewett, D.T., Cahill, J., Chabot, N., Greenhagen, B., Hibbitts, C., Klima, R., Lawrence, D., Mandt, K., Nunez, J.I. and Patterson, W., Prem, P., Stickle, Angela., Deutsh, Ariel., Delitsky, M., Flahaut, J., Sowers, G., Sefton-Nash, E., Brisset, J., Zacny, K., Shukla, S., Grandidier, J., Hosseini, S., Liu, Y., Bhiravarasu, S., Stopar, J., Barker, D., Gertsch, L., Jha, D., Colaprete, A., Elphic, R., Banks, M., Benna, M., Farrell, W., Saena, P., Tucker, O.J., Mitchell, J., Needham, D., Atkinson, J., Barber, S., Hendrix, A., Kramer, G., Siegler, M., Grava, C., Poston, M., Retherford, K., **Kumari, N**., Barnes, J., Hayne, P., Li, S., Lucey, P., Livengood, T., 2021. Mission to Characterize Volatiles in Old, Cold, Permanently Shadowed Regions on the Moon. *Bulletin of the American Astronomical Society*, *53*(4), p.365.

[1] Sinha, R.K., Sivaprahasam, V., Bhatt, M., Harish, **Kumari, N.**, Srivastava, N., Varatharajan, I., Ray, D., Wöhler, C. and Bhardwaj, A., 2020. Geological Characterization of Chandrayaan-2 Landing Site in the Southern High Latitudes of the Moon. *Icarus*, *337*, p.113449.

MANUSCRIPT IN PROGRESS

[17] **Kumari, N.,** Breitenfeld, L. B. (2026). Quantifying Shock Pressures Experienced by Feldspars Using Spectroscopy. (*Manuscript in Preparation*)

[16] Luo, R.*, **Kumari, N.*,** Mustard, J.F. (2026). Investigating the Formation of Late-Stage Shield Volcanoes on the Moon: A Case Study of the Struve-Russell Complex. (*With co-authors*)

[15] **Kumari, N.,** Head, J.W. (2026). Lunar Laccoliths as Revealed by Impact Crater Floor Doming: A New Class of Shallow Magmatic Lunar Basalt Intrusions and Implications for Magma Ascent and Eruption Dynamics. (*Manuscript in Preparation*)

[14] **Kumari, N.,** Gary-Bicas, C., Ye, C., Rogers, A.D., Glotch, T. D., Mustard, J. F., (2026). Decoding Felsic Lithologies on the Rocky Worlds: The Silica Story (*With co-authors*)

FIRST AUTHORED CONFERENCE ABSTRACTS

[17] **Kumari, N.,** Robertson, K., Milliken, R. E., Mustard, J. F., (2025). Spectral Characterization of the Lunar Meteorite Northwest Africa (NWA) 11444, *LPSC*

[16] **Kumari, N.,** Williams, J. P, Bhatt, H., Glotch, T. D. , Jhoti, E., Powell, T. (2024). Probing South Polar Region, *NASA Exploration Science Forum*

[15] **Kumari, N.,** Williams, J. P, Glotch, T. D. , Jhoti, E., Powell, T., Bhatt, H. (2024). Christiansen Feature Maps of the Lunar Poles, *LPSC #1819*

[14] **Kumari, N.,** Glotch, T. D. , Greenhagen, B. T., Breitenfeld, L. B. (2023). Bulk Composition Characterization of the Lunar Surface Using Machine Learning: Applications to Ongoing and Upcoming Missions, *NASA Exploration Science Forum*

[13] **Kumari, N.,** Glotch, T. D. , Greenhagen, B. T., Breitenfeld, L. B. (2023). Brewing Machine Learning Models for Accurate Analysis of Current and Future Datasets, *LPSC #2396*

[12] **Kumari, N.,** Prettyman, T. H. , Lane, M., Martin, A. C., Patterson, R. V., Meier, M. L., Aherns, C. J., Pearson, N. C., Clark, R. N., Vilas, F., Steckel, A. V., Knightly, J. P., Wettergreen, D., Banks, M. E., Bell, E., Wright, S. P., Noe Dobrea., E. Z., Hendrix, A., (2023). Procedure Standardization for Toolbox for Research and Exploration Field Data Analysis, *LPSC #1814*

[11] **Kumari, N.,** Glotch, T. D. , Williams, J. P, Greenhagen, B. T., Li, S., Waller D., Shirley, K. A., Sullivan, M. T. (2022). Mineralogical and Thermophysical Properties of Gruithuisen Domes: Preparations Towards Landing, *International Mineralogical Association*

[10] **Kumari, N.,** Glotch, T. D. , Williams, J. P, Greenhagen, B. T., Li, S., Waller D., Shirley, K. A., Sullivan, M. T. (2022). Gruithuisen Domes: Silicic Volcanic Constructs on the Moon, *Lunar Exploration Analysis Group*

[9] **Kumari, N.,** Glotch, T. D. , Williams, J. P, Greenhagen, B. T., Sullivan, M. T. (2022). A Holistic View of Gruithuisen Domes Silicic – Constructs on the Moon, *LPSC #2034*

[8] **Kumari, N.,** Bretzfelder, J.M., Ganesh, I., Lang, A., Kring, D.A., (2021). Surface Illumination, Temperatures and Cold Traps at Two Potential Landing Sites near the Lunar South Pole, *NASA Exploration Science Forum*

[7] **Kumari, N.,** Bretzfelder, J.M., Ganesh, I., Lang, A., Kring, D.A., (2021). Geological Diversity at Two Potential Landing Sites in the Lunar South Pole, *LPSC #1197*

[6] Kring, D.A., Bretzfelder, J.M., Ganesh, I., **Kumari, N.,** Lang, A. (2021). Artemis III EVA Opportunities on the Rim of De Gerlache Crater, *NASA Requested Input for Artemis III SDT*

[5] Kring, D.A., Bretzfelder, J.M., Ganesh, I., **Kumari, N.,** Lang, A. (2021). Alternate Artemis III EVA Near De Gerlache Crater, *NASA Requested Input for Artemis III SDT*

[4] Kring, D.A., Bretzfelder, J.M., Ganesh, I., **Kumari, N.,** Lang, A. (2021). Artemis III EVA Opportunities on the Lunar Farside Near Shackleton Crater, *NASA Requested Input for Artemis III SDT*

[3] **Kumari, N.,** Glotch, T. D. , Greenhagen, B. T., Shirley, K. A.,(2020). Different Trends of Variation in Christiansen Feature on the Lunar Surface: Effects of Iron Content and Particle Size, *NASA Exploration Science Forum*

[2] **Kumari, N.,** Glotch, T. D. , Greenhagen, B. T., Shirley, K. A.,(2020). Investigation of Variation in Christiansen Feature with respect to Albedo on the Moon using the Diviner, M^3 and Kaguya Datasets, *LPSC #1967*

[1] **Kumari, N.,** Vijayan, S. (2019). Global Distribution and Characteristics of Domical Craters on the Moon, *LPSC #1803*

TECHNICAL SKILLS & MISSION EXPERIENCE

PROGRAMMING	Python(Spatial and Non-spatial Data Analysis/Machine Learning, Proficient)
INSTRUMENTS	FTIR, Micro-FTIR, ASD, SEM(BSE and EDS mapping), EPMA
SOFTWARE	ArcGIS, QGIS, ENVI, Bruker FTIR softwares, ERDAS Imagine, ISIS, Davinci
DATA PROFICIENCY	Earth: LANDSAT[3] (VIS, THERMAL, PAN), ASTER Thermal InfraRed Data; Moon: LRO (WAC, NAC, WAC-DTM, Diviner, LOLA), Chandrayaan-1(M^3), Kaguya SELENE(MI, TC) ; Mars: MRO (CRISM, CTX, HiRISE), MGS(MOLA), Mars Odyssey (THEMIS)

[3] LANDSAT Data from 5 to 8

FIELD WORK

KILBOURNE AND HUNTS HOLE, SSERVI (RISE4) 2022;2023
SAN FRANCISCO VOLCANIC FIELD, SSERVI (CLSE) 2022
UTAH YELLOWCAT SITE, SSERVI(TREX),2022
MOUNT ASCUTNY, *NEW ENGLAND NEIGC FIELD TRIP 2019, 2022*

LEADERSHIP SKILLS

SENATOR	| GRADUATE STUDENT ORGANIZATION OF SBU	[2021-2023]
STUDENT DIRECTOR	| *CEG TECH FORUM*	[2014-2018]

PROFESSIONAL EXPERIENCES & SERVICES

CHAIR /MODERATOR	| *LPSC, 2024, 2026, LEAG 2022, NESF 2020,2024*	
REVIEWER	| *JGR:PLANETS, ICARUS, PSJ, GRL, MAPS, PRJ*	
REVIEWER	| *NASA PANEL 2023, NASA PANEL 2026*	
EXECUTIVE SECRETARY	| *NASA PANEL 2022, 2024, 2025*	
SUMMER INTERN	| *LUNAR AND PLANETARY INSTITUTE*	[MAY'20-AUG'20]
VISITING STUDENT	| *PHYSICAL RESEARCH LABORATORY*	[MAY'18-JUL'19]